

1.What is RDD?

Apache Spark **RDD (Resilient Distributed Dataset)** is Spark's fundamental data structure used for distributed and parallel data processing.

Features

- Distributed across multiple machines
- Immutable
- Fault tolerant
- Supports parallel processing

2.Advantages of RDD

Advantage	Description
Fault Tolerance	Recovers lost data using lineage
Parallel Processing	Multiple nodes process data simultaneously
Scalability	Can handle very large datasets
In-memory Processing	Faster computation
Lazy Evaluation	Optimizes execution before running

3.What is Lazy Evaluation?

In Spark, transformations are not executed immediately.

Execution starts only when an **action** is called.

Example:

```
rdd.map(lambda x: x * 2) # Transformation
rdd.collect()           # Action
```

Benefits

- Optimized execution using DAG
- Reduces unnecessary computation
- Improves performance